

Rebecca Karungi

Full-Stack Software Developer

Kigali, Rwanda | [LinkedIn](#) | [GitHub](#) | karbecca3@gmail.com | +250 790 774 983

SUMMARY

Full-Stack Software Developer with expertise in building scalable, user-centered web applications and RESTful APIs using React, TypeScript, Python (FastAPI), Java (Spring Boot), Node.js (NestJS), and SQL/NoSQL databases. Proven track record of developing end-to-end systems with secure authentication, role-based access control, and modern cloud/containerized deployment. Adept at collaborating in SCRUM environments, writing clean tested code, and delivering high-performance solutions.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, Python, Java, HTML5, CSS3
- **Frontend:** React, Next.js, Redux, Tailwind CSS
- **Backend & APIs:** NestJS, Node.js, FastAPI, Spring Boot, RESTful APIs
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Cloud & DevOps:** Docker, DigitalOcean, Git, GitHub
- **Quality & Security:** Jest, Cypress, JWT, 2FA/OTP Authentication, Audit Logging
- **Methodology & Tools:** Agile/SCRUM, Figma, Trello, Code Reviews

WORK & TRAINING EXPERIENCE

Specialized Engineer Trainee – Backend & AI Systems

The Gym Rwanda — Kigali, Rwanda | February 2026 – Present

- Architect and develop scalable RESTful microservices using Python (FastAPI), integrating Kafka for real-time data streaming within distributed backend systems.
- Engineer enterprise-grade authentication flows, containerized deployments with Docker, and production-ready security practices across backend services.
- Apply clean architecture patterns (layered design, dependency injection) and write automated tests with Jest to ensure code quality and long-term maintainability.

Software Development Trainee (Phase 1)

The Gym Rwanda — Kigali, Rwanda | November 2024 – January 2026

- Delivered multiple full-stack applications across the complete development lifecycle — from React/TypeScript frontends and NestJS/PostgreSQL backends to containerized deployment with Docker.
- Engineered a car-sharing web platform in React and NestJS enabling users to list, search, and book vehicles, with PostgreSQL-backed rental management and persistent user sessions.
- Collaborated in SCRUM agile teams, utilizing Figma for UI prototyping and Git/GitHub for version control and peer code reviews.

FULL-STACK PROJECTS

AI-Based Patient Flow & Waiting Time Prediction Platform (Final-Year Project)

- Engineered a full-stack hospital operations platform featuring React/TypeScript role-based dashboards with clinical urgency scoring (L1–L5) and a Python/FastAPI REST backend with PostgreSQL, designed for integration with existing Electronic Medical Record systems.
- Trained Random Forest models for patient wait-time prediction and 24-hour demand forecasting, achieving a Mean Absolute Error of **±12.64 minutes** on validation data.
- Ensured application security with JWT authentication, email-OTP 2FA, role-based access control, and comprehensive system audit logging.

AI Vacation Planner (Full-Stack Capstone Project)

- Built an end-to-end web assistant in Python, FastAPI, and React with JWT authentication, CRUD operations for trip planning and user management, and Swagger-documented REST endpoints.
- Integrating LLM APIs (chat completions and streaming) to power natural-language itinerary generation, enabling conversational end-to-end trip planning.

VOLUNTEER EXPERIENCE

Volunteer Coach

The Gym Rwanda — Kigali, Rwanda | April 2025 – September 2025

- Coached cohorts of junior developers across preparatory sprints in JavaScript, DOM manipulation, frontend/backend fundamentals, and algorithmic problem-solving.
- Conducted peer code reviews and provided technical feedback, reinforcing collaborative development practices and code quality standards.

EDUCATION

Honours Bachelor of Software Engineering

Adventist University of Central Africa — Kigali, Rwanda | 2023 – 2026

- Expected graduation: 2026. Focus areas: Software Architecture, Full-Stack Engineering, Database Systems, and Algorithms.

CERTIFICATIONS

- **Certificate of Graduation – The Gym Rwanda (January 2026)** — Full-Stack Software Development
- **Advanced Network Operations Certificate** — Internet Society
- **CISCO Network Essentials Certificate**